YZM5110

Multimodal Learning

Week 1: Foundations and Readiness Recap

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Course Philosophy

Principle	What it means here
Reasoning over recall	The final exam is open-book, open-web, and open-AI
Research and practice	Two individual projects, no midterm
Autonomy	Attendance is not graded
Sustainable attention	Short focused blocks

These principles also define the work expected in the course.

- When you answer a question or present a result, state your assumptions, explain your design choices, and connect each claim to evidence.

Assessment

Component	Weight	Deadline	What it tests
Research project	30%	Week 7	Critical literature evaluation, reproduction, and an improvement study
Practical project	30%	Week 12	An original multimodal solution
Final examination	40%	Date to be announced	Full-course understanding and system design

The Research Project

Due Week 7. Individual submission. 30% of the course grade.

Choose a focused multimodal-learning topic and gather a coherent set of papers.

- Develop an in-depth evaluation of their questions, assumptions, methods, datasets, baselines, results, limitations, and reproducibility.
- Explain agreements and contradictions across papers rather than presenting disconnected summaries.

Implement or adapt code to run a central concept, reproduce a specific published result at a feasible scale, and evaluate it with fair controls.

- Then identify one weakness, implement a targeted improvement, and measure its effect against the reproduced baseline. Report unsuccessful improvements honestly.

Submit a detailed literature and experimental report, a comparison matrix, a runnable repository with results, and a presentation defending your analysis.

- A reading-only survey or an unchanged demo is incomplete.
- Include a Hugging Face Spaces demo or a GitHub repository with runnable demonstration instructions.

The Practical Project

Due Week 12. Individual submission. 30% of the course grade.

Tackle a concrete problem using at least two modalities.

- State who needs the solution, what evidence each modality contributes, and how success will be measured.
- Build and evaluate a working system with unimodal baselines and a test of missing, corrupted, or conflicting inputs.

Your contribution must be **novel**.

- Compare it with the closest papers and existing systems, then identify the specific new problem formulation, method, interaction design, or evaluated capability you contribute.
- A renamed tutorial, an API wrapper, or a dataset change without a justified new contribution is insufficient.

Novelty does not require a new foundation model or a state-of-the-art score.

- It requires an original, defensible contribution supported by a working implementation and evidence.

Include a Hugging Face Spaces demo or a GitHub repository with runnable demonstration instructions.

Project Authorship and AI Assistance

Research project	Practical project
Disclosed AI assistance for search, coding, or editing is permitted.	AI can be used to write submitted code, text, figures, slides, or analysis. But it should NOT be used for ideation.
Read the original sources, understand the code, run the experiments, and verify every claim yourself.	Develop the project-specific work yourself. Coding agents can help you but you should OWN decisions.

Keep meaningful Git history, design notes, experiment logs, and a reused-component inventory. Be ready to explain your decisions and modify your code. Identify reuse between the two projects; the same work cannot receive credit twice.

Two Projects, Two Distinct Contributions

Submission	Required evidence	Due
Research	Deep paper comparison, reproducible experiment, evaluated improvement, report and presentation	Week 7
Practical	Novel problem solution, human-authored implementation, baselines, robustness tests, report and demonstration	Week 12

Both projects are individual. You may develop a related topic across them, but identify separate questions, contributions, experiments, and submissions.

- Cite any reused work from the first project.

Start the research project now with papers and a small executable baseline.

- Develop the practical idea alongside it so there is time to test novelty and data access.

Questions?

What Is Multimodal Learning?

Consider a bell:

- A photograph shows its shape.
- Audio can reveal that it rang.
- Written context can explain why it was rung.

These are **partial observations**, not interchangeable copies of reality. Whether combining them helps depends on the question.

Multimodal learning therefore requires a task-specific account of which information each observation contributes and how their relationship is learned.

Source: Baltrusaitis, Ahuja, and Morency - Multimodal Machine Learning: A Survey and Taxonomy

What Is Multimodal Learning?

Multimodal learning studies models that learn from and connect information expressed in more than one modality.

> The central problem is not just receiving multiple inputs. It is learning what each source preserves, how the sources correspond, and when their interaction helps the task.

Examples include answering a question about an image, retrieving a video from a sentence, and recognizing speech from sound plus lip movement.

Source: Baltrusaitis, Ahuja, and Morency - Multimodal Machine Learning: A Survey and Taxonomy

Why We Study Multimodal Learning

Modern AI systems rarely receive a single, neatly organized table of numbers.

They receive **language, pixels, sound, motion, documents, sensor streams, and actions** - often incomplete, asynchronous, noisy, or contradictory.

Each source organizes information differently. As we work through the course, we will keep asking four practical questions:

1. What should be preserved?
2. How should observations be connected?
3. When should they interact?
4. What evidence would show that the combination helped?

A Question to Carry Through the Course

What evidence would show that a model used more than one modality for the right reason?

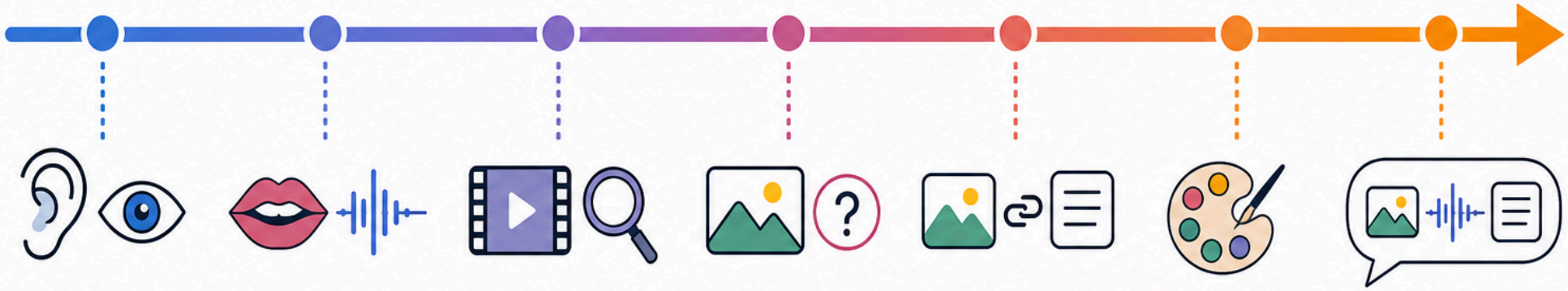
A high score is not enough:

- perhaps text alone predicts the label;
- perhaps one modality dominates training;
- perhaps the benchmark leaked into pretraining;
- perhaps the explanation is fluent but visually ungrounded.

Throughout the course, we will treat multimodal capability as an **empirical claim**.

We will test it by removing or corrupting a modality, changing the evidence counterfactually, and comparing against strong unimodal baselines.

A Brief History of Multimodal Machine Learning



1970s-80s: McGurk Effect & audio-visual speech

1990s: Audio-Visual Speech Recognition

2000s: Multimedia retrieval & affect recognition

2015: Image captioning & VQA

2021: CLIP - contrastive vision-language pretraining

2022: Stable Diffusion & Flamingo

2023-now: GPT-4o class native multimodal models

Block 1 - Modalities as Partial Observations

> If the sources differ in structure, noise, timing, or semantic role, what must the model do differently for each one?

What Is a Modality?

A modality is a way in which information is **expressed or perceived**, with characteristic structure and noise.

Examples include language, image, audio, video, touch, depth, IMU, EEG, tables, graphs, and action.

The boundary here is contextual. For example, RGB and depth may be separate sensor modalities in robotics but two channels of one visual observation in another study.

For this course, a modality is an **operational modeling choice**. We call two sources different modalities when their structure, noise, sampling process, or semantic role requires meaningfully different representation or evaluation decisions. The definition should help us design the system.

Modality, Model Architecture, and Output Contract

A modality describes the kind of information available, such as an image, speech, or text. Architecture describes how a model processes that information. The output contract specifies what software receives: a sentence, a class, a score, or a probability distribution. Changing one does not automatically change the others.

Question	Example
What evidence enters?	A document image and a written request
How is it processed?	A visual encoder, text encoder, and fusion module
What must come out?	A routing decision with probabilities

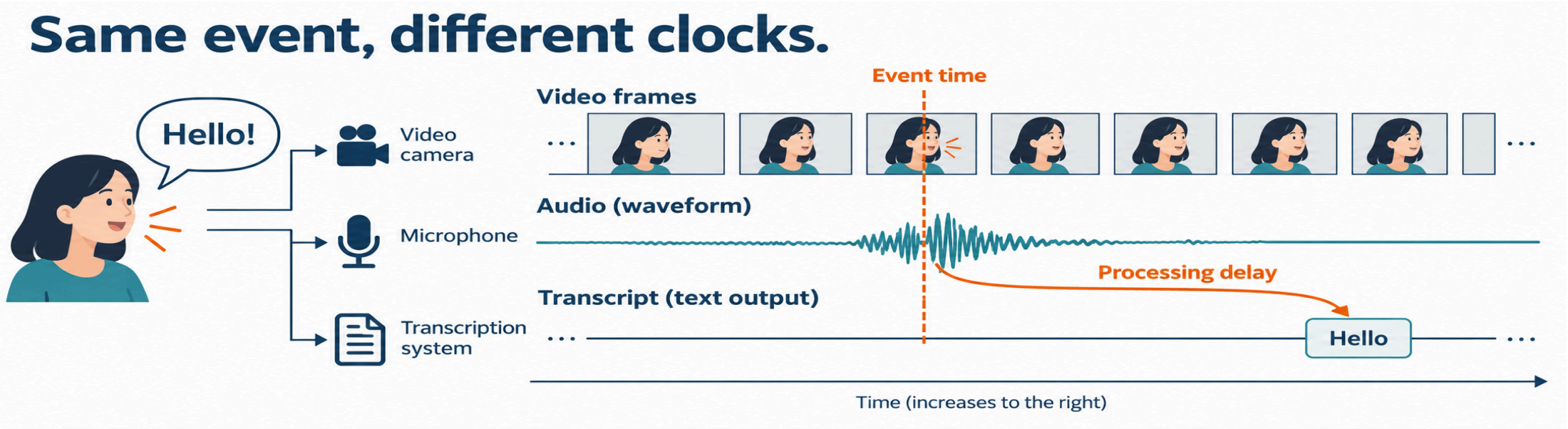
Recent example: Jev returns typed decisions from text input. It does not currently accept images, audio, or video. It could consume text extracted by a separate perception model, but that would make the overall pipeline multimodal, not Jev itself.

Modality, Channel, View, Sensor

Term	Useful distinction
Modality	form and structure of information
Channel	transmission or component within a modality
View	perspective on the same underlying event
Sensor	physical or software measurement device

For example, a stereo camera provides two views through two sensors, but both observations may use the same visual modality and encoder.

A document page and its OCR text may come from one physical source yet behave as distinct modalities because one preserves layout and appearance while the other provides discrete language tokens.

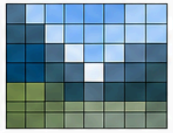
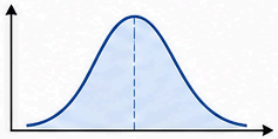
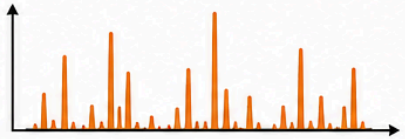
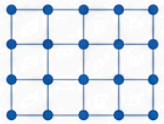


A meeting camera might record **30 frames per second**, audio thousands of samples per second, and a transcript words or sentences with processing delay.

You should not confuse **when an event happened** with **when its transcript became available**. Motion blur, occlusion, background speech, and recognition errors add uncertainty.

Tracing "hello": Which video frames correspond to the audio event, not the later transcript arrival? Resampling, alignment, or tolerance windows may be needed. It's a good idea to record modality availability and keep related recordings together when splitting data.

Dimensions of Modality Heterogeneity

	Image Modality	**Text Modality**
1 **Element representation**	 Pixels (grid of numeric values)	the cat sat on the mat ... Discrete tokens (sequence of words)
2 **Distribution**	 Dense continuous (bell-shaped)	 Sparse symbolic (spiky)
3 **Structure**	 2D spatial grid (height × width)	1D sequence (order matters)
4 **Information**	Varying amounts per element (e.g., intensity, color)	Varying amounts per token (e.g., informativeness, rarity)
5 **Noise**	Blur, occlusion, lighting changes, sensor noise	The ct sat on the mat. Typos, misspellings, missing words, grammatical errors
6 **Relevance**	Which modality is more relevant for the task?	Which modality is more relevant for the task?

Heterogeneity - Element Representation

Element representation concerns the basic units in which a modality is expressed.

- Text uses discrete symbols, images contain spatially arranged pixel measurements, and audio is a rapidly sampled waveform.
- Before these observations can interact, the model must convert them into features that retain the distinctions required by the task.
- Mapping everything to vectors provides a common computational interface, but it does not remove the differences in what the units mean.

Tokenization creates comparable interfaces while preserving different inductive biases.

An image patch, a wordpiece, and an audio frame may all become vectors, but their neighborhoods mean different things. Convolution, temporal masking, relative position, and graph message passing encode assumptions about which local relations are likely to matter.

Heterogeneity - Distribution and Structure

Distributions differ in scale, sparsity, entropy, and sampling frequency.

Structures differ in topology:

- sequences have order;
- images have locality and translation structure;
- sets are permutation invariant;
- graphs have relational neighborhoods;
- video combines space and time.

These structures determine permissible transformations.

- Reordering words usually changes meaning, reordering a set should not
- Translating an object within an image may preserve its identity, and shifting audio slightly in time may or may not preserve alignment with video.

A model’s inductive bias should respect these symmetries.

Heterogeneity - Information

Two modalities can refer to the same event while revealing different facts about it.

- A photograph can reveal color but cannot directly supply a sound.
- The absence of a detail in one modality is not necessarily an error.

Let X_1 and X_2 be modalities and Y the target.

- **Redundant:** both explain overlapping information about Y
- **Unique:** only one explains a component of Y
- **Synergistic:** the combination explains what neither alone can

Fusion should preserve the information the task needs and make that information available at the right stage.

Block 2 - Representations Preserve Evidence

> What information does the system preserve before it starts making predictions?

Recap - Models Consume Tensors

A sample becomes a tensor with **values, axes, and semantics**:

Input	Example shape	Axes mean
Text tokens	L	sequence position
Image	$H \times W \times C$	two spatial axes and channels
Audio	T or $F \times T$	samples, or frequency over time
Video	$T \times H \times W \times C$	time and space

The shape is not the representation. The architecture decides what relationships the axes permit.

A batch axis has been omitted for clarity.

Axis meaning determines which augmentations, positional encodings, normalization operations, and attention patterns are sensible.

- Two tensors can have identical dimensions while describing entirely different phenomena.

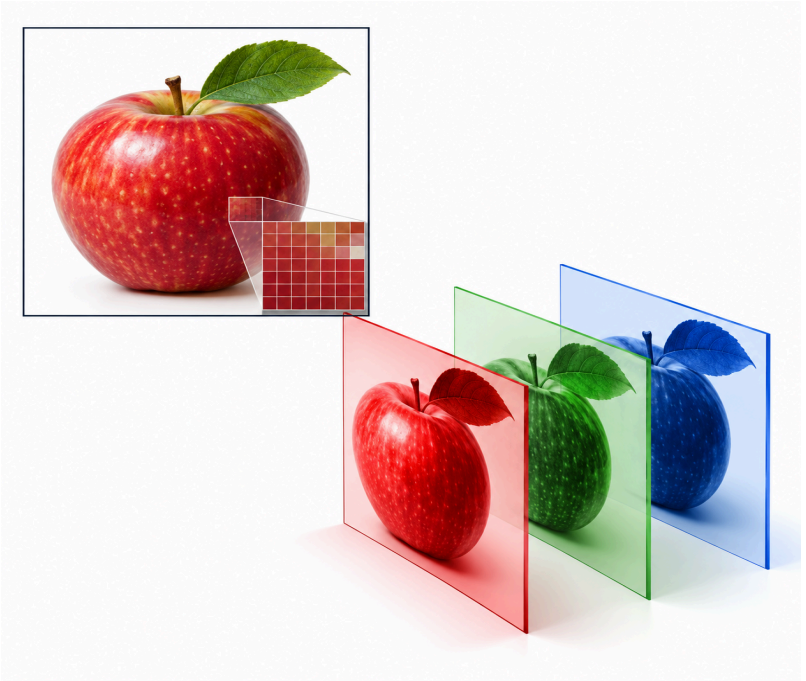
A Photograph Becomes a Tensor

Each tensor axis answers a different question: **which row, which column, which color channel?**

For a 224×224 RGB photograph:

1. Pixel $(80, 120)$ stores red, green, and blue intensity at the **same location**.
2. Neighboring rows and columns usually describe neighboring parts of the scene.
3. Convolution uses this local-neighborhood assumption.
4. Flattening gives $150{,}528$ values but hides their spatial organization unless we retain that mapping.

Core idea: Tensor shape records arrangement and the model architecture decides how that arrangement may be used.



Recap - Representation Learning

A representation is a numerical description that makes selected properties of an observation available to a model.

- The encoder is the function that creates it, and its parameters determine which distinctions become easy to express.
- Similar representations are useful only when similarity corresponds to something relevant to the task.
- For example, recognizing an animal may benefit from ignoring its background, while locating that animal requires spatial detail.

We learn a function

$$f_\theta : \mathcal{X} \to \mathbb{R}^d$$

that maps a complex observation to features useful for a family of tasks.

Good representations usually make relevant factors:

- easier to separate or retrieve;
- stable under irrelevant variation;
- sensitive to changes the task actually cares about.

Representation Is a Contract

An embedding vector has no universal interpretation.

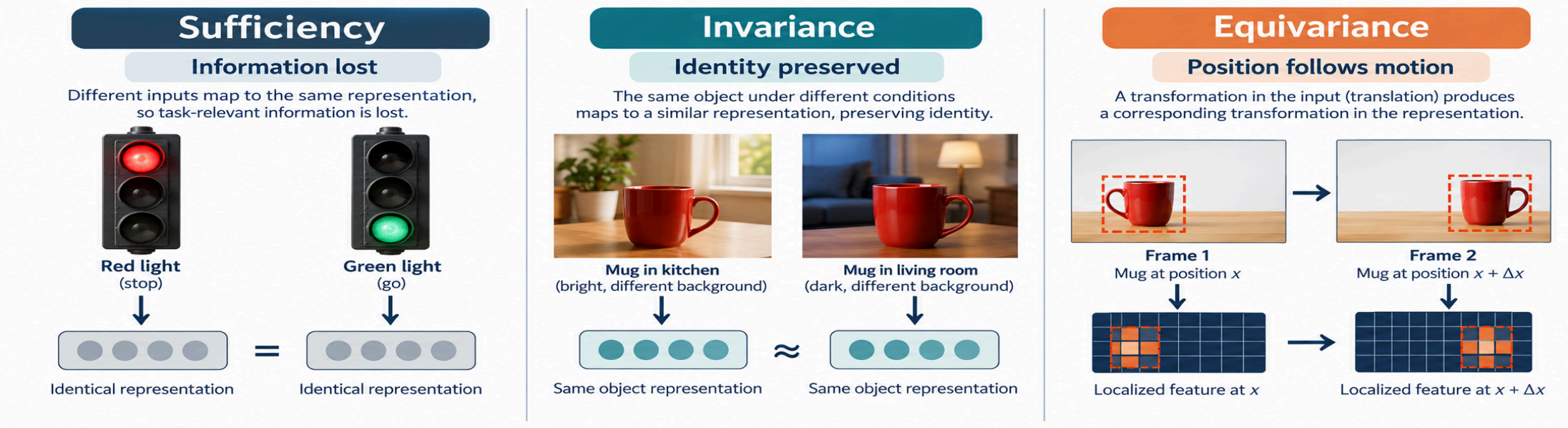
Its geometry is shaped by:

- the data distribution;
- the objective and positive/negative pairs;
- the architecture's inductive biases;
- the downstream task and evaluation.

Never say "the representation is better" without specifying **for what** and **under which probe**

For example, an embedding that discards color may be excellent for shape-based retrieval and unusable for traffic-light recognition. Representation quality is therefore relational: it depends on the information retained and the decision we later ask the representation to support.

Three Tests for a Representation



- **Sufficiency:** is the evidence needed for the task still available?
- **Invariance:** does an irrelevant change leave the relevant feature stable?
- **Equivariance:** does a meaningful transformation produce a corresponding feature change?

Analogy - Every Map Leaves Something Out

A road map, metro map, and weather map can describe the same city, yet each preserves different relationships.

Map	Retains for its task	Can omit
Road	Street connectivity for driving	Building interiors
Metro	Station order and transfers	Exact geography
Weather	Temperature and pressure	Street names

An embedding is similarly **shaped by its task**. Compression helps until it removes information the next task needs.

For example, grouping red and green traffic lights by shape may help object recognition but discard evidence needed for driving.

Which map would you choose to change trains? To avoid a storm?

Question - Preserving What the Task Needs

You are designing a model for **“Is the pedestrian wearing a reflective vest?”**

Which representation choice is most risky?

- A global vector that removes local spatial detail
- A patch representation that retains color and position
- A feature that is stable to small lighting changes
- A representation evaluated with a location-aware probe

Question - Preserving What the Task Needs

You are designing a model for **“Is the pedestrian wearing a reflective vest?”**

Which representation choice is most risky?

- A global vector that removes local spatial detail

- A patch representation that retains color and position
- A feature that is stable to small lighting changes
- A representation evaluated with a location-aware probe

The first choice is risky because the answer depends on a local visual attribute. You should always connect a representation decision to the evidence required by the task.

Locating the evidence: Point to the clothing detail the representation must preserve. Recognizing “a pedestrian” alone does not answer the question.

Block 3 - Learning Signals and Responsibility

> Which behavior does the loss reward, and which component receives responsibility for the error?

Recap - Three Supervision Regimes

Regime	Training signal	Multimodal example
Supervised	Human-provided target y	emotion label from video and speech
Self-supervised	Structure inside the data	masked audio prediction
Weakly supervised	Naturally occurring noisy relation	image paired with alt text

Foundation models often mix all three across pretraining and post-training.

The regimes are distinguished by the **source of the learning signal**, not by the model architecture.

- A single model may learn from web-scale weak pairs, self-supervised masking, and a smaller supervised instruction dataset.
- Each signal introduces different assumptions and characteristic label noise.

An everyday analogy is learning with three kinds of feedback: an instructor provides the answer (**supervised**), a puzzle hides part of its own content (**self-supervised**), or a noisy web label gives an imperfect hint (**weakly supervised**).

Recap - Learning as Optimization

Training searches for parameters that minimize expected loss:

$$\theta^* = \arg\min_{\theta} \; \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[\mathcal{L}(f_\theta(x), y)\right]$$

Though, this equation hides the decisions that matter:

- what counts as x and y;
- how the dataset $\mathcal{D}$ was produced;
- which errors the loss rewards or ignores;
- whether train and deployment distributions match.

In practice, we minimize an empirical average over a finite training set, not the unknown deployment expectation.

Generalization depends on whether this sample preserves the relevant relationships among modalities. Dataset construction is therefore part of the learning algorithm, not just a preprocessing step.

Think of the loss as a **scoring rule** and the optimizer as a navigator following local slope information. If the scoring rule rewards the wrong behavior, more efficient navigation reaches the wrong destination faster.

When the Loss Rewards a Shortcut

Suppose training images labelled **chair** are mostly indoors. Both chair shape and indoor background may reduce training loss.

The loss rewards agreement with targets on the examples provided. It does not know which visual evidence we intended.

Read the three scenes:

1. Indoor chair: both strategies predict “chair.”
2. Outdoor chair: a background shortcut may fail.
3. Empty room: an indoor-background rule may give a false positive.

To Test: Vary object presence and setting separately. In multimodal data, text metadata can become an even easier shortcut.

Recap - The Training Loop

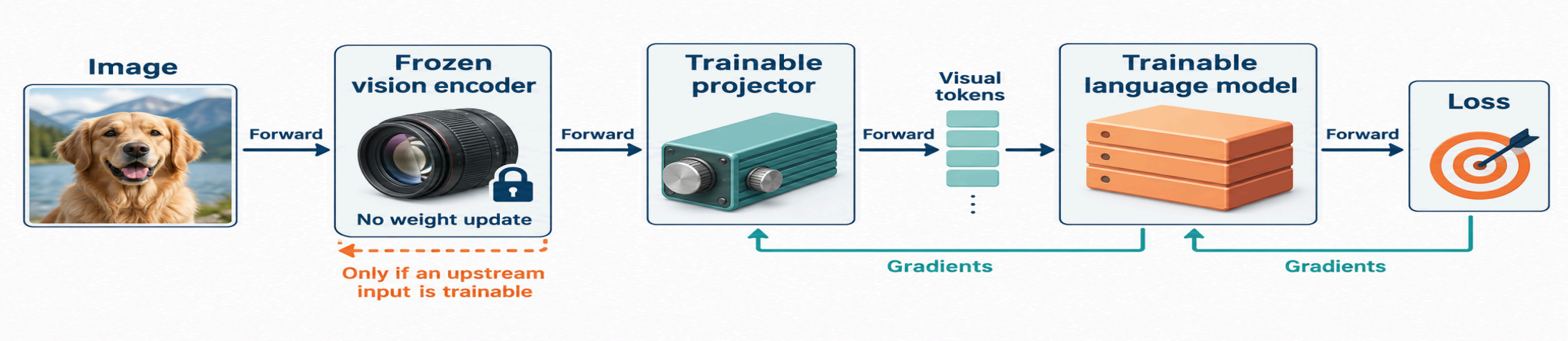
```
for batch in loader:
    prediction = model(batch.inputs)   # forward
    loss = objective(prediction, batch.targets)
    optimizer.zero_grad()
    loss.backward()                    # gradients
    optimizer.step()                   # parameter update
```

The multimodal complication is deciding which encoders receive gradients, how losses are balanced, and whether one modality learns faster than another.

Backpropagation follows every differentiable path from the loss to a trainable parameter. With several objectives - captioning, contrastive alignment, reconstruction, or instruction following - their scales and weights determine which behaviors dominate the update.

In a multimodal model, gradients are the assignment of **responsibility**. They indicate which adjustable components contributed to the error and in which direction those components should change.

Gradient Path Audit



Trace **forward evidence**, **backward gradients**, and **parameter updates** separately.

1. **Which weights can change?** The projector and language model.
2. **Does a frozen encoder ignore the image?** No. Its weights stay fixed, but its activations depend on the input.
3. **Would detaching its output block an earlier trainable module?** Yes. Detaching breaks this loss's gradient path to that upstream module. Downstream modules can still learn.

Freezing weights and detaching activations are different operations. To train an upstream component through a frozen encoder, retain the differentiable path.

Source: PyTorch - Autograd mechanics and Tensor.detach

Frozen Does Not Mean Irrelevant

Freezing a parameter prevents its update but does not make the component's output constant.

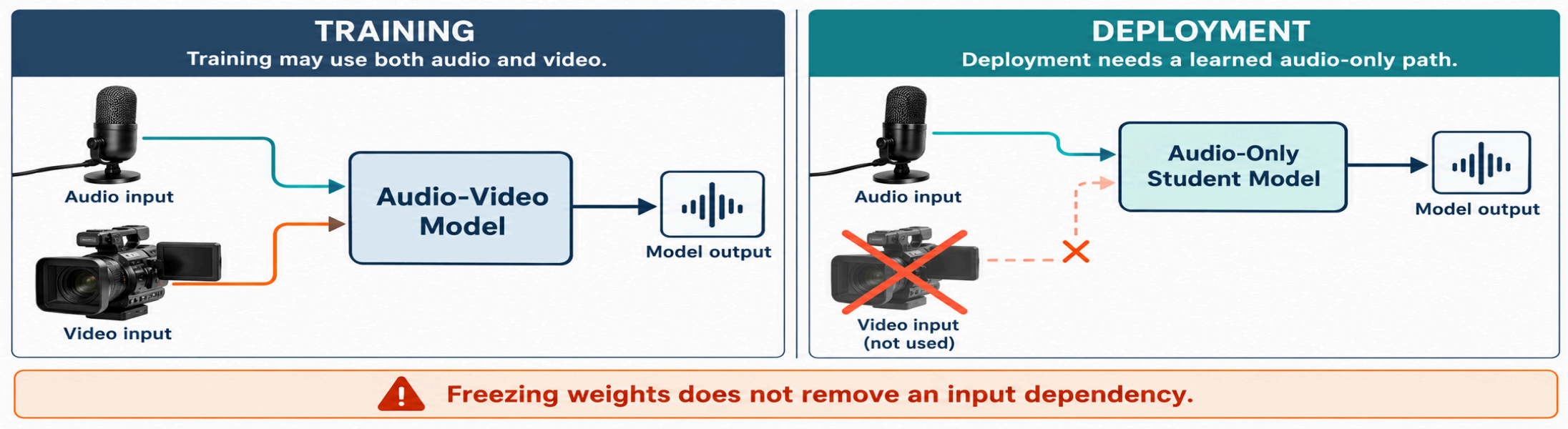
If a pretrained vision encoder is frozen:

- its weights stay fixed;
- its features still change with the image;
- a trainable connector or language model can learn to use those features;
- gradients may pass through it when an earlier trainable component needs them.

If there is no upstream component to train, the encoder can provide features without retaining its backward graph. If an upstream component must learn through it, detaching its output would block that path.

Freezing trades **adaptability** for **stability** and **efficiency**. It can reduce training memory and compute and preserve a useful pretrained representation. The saving depends on which gradient paths are still required.

Training Evidence and Deployment Inputs



A speech model learns from paired audio, video, and task targets. The deployed device has **only a microphone**.

A frozen vision encoder still needs an image to produce visual features. Freezing does not make its input optional.

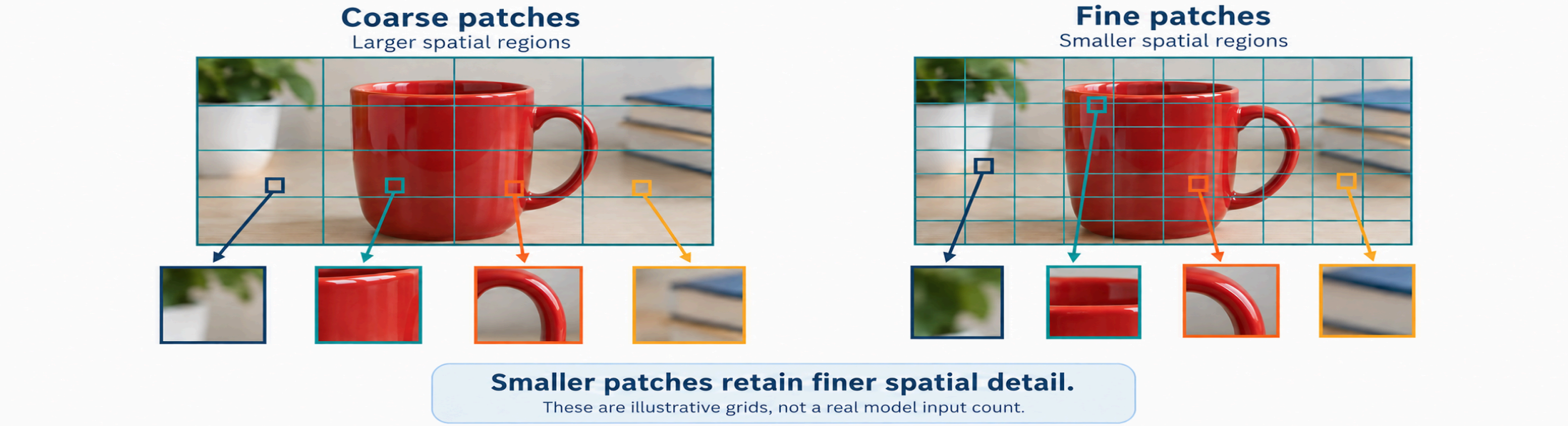
If the deployment still calls the vision encoder, what must change?

- The design must support audio-only prediction, for example through an explicitly trained student or missing-modality strategy. We will study those methods later.

Block 4 - Tokens and Attention as Interfaces

> What becomes visible to attention, and what has already been compressed away?

Recap - From Objects to Tokens



A token is an **addressable unit of computation**: text subwords, image patches or regions, audio frames or codec codes, video tubelets, or action bins and vectors.

Smaller image patches can preserve finer detail but create more tokens. Large patches may mix several objects before attention sees them. Coarse audio frames blur timing; rare text fragments can require more subwords.

Tokenization determines what the model can address and may discard evidence that later layers cannot recover.

Embeddings Turn Tokens into Vectors

For token ID i, an embedding table returns row $E_i \in \mathbb{R}^d$.

For continuous inputs, a learned projection serves the same role:

$$z_j = Wx_j + b$$

Different modalities can enter one Transformer only after they agree on the model dimension d - but equal dimension does **not** imply equal meaning.

A projection aligns the **interface**, not necessarily the semantic space.

- If image and text vectors should be directly comparable, the training objective must shape their geometry accordingly.
- Otherwise, equal-sized vectors are simply compatible inputs to later layers.

Tokens Need Position and Time

Token content alone does not tell the model **where or when** it occurred.

Modality	Missing relationship	Position information
Text	“dog bites person” versus “person bites dog”	sequence position
Image	red mug left of versus right of the book	patch row and column
Audio/video	a sound before versus after an action	timestamp or temporal position

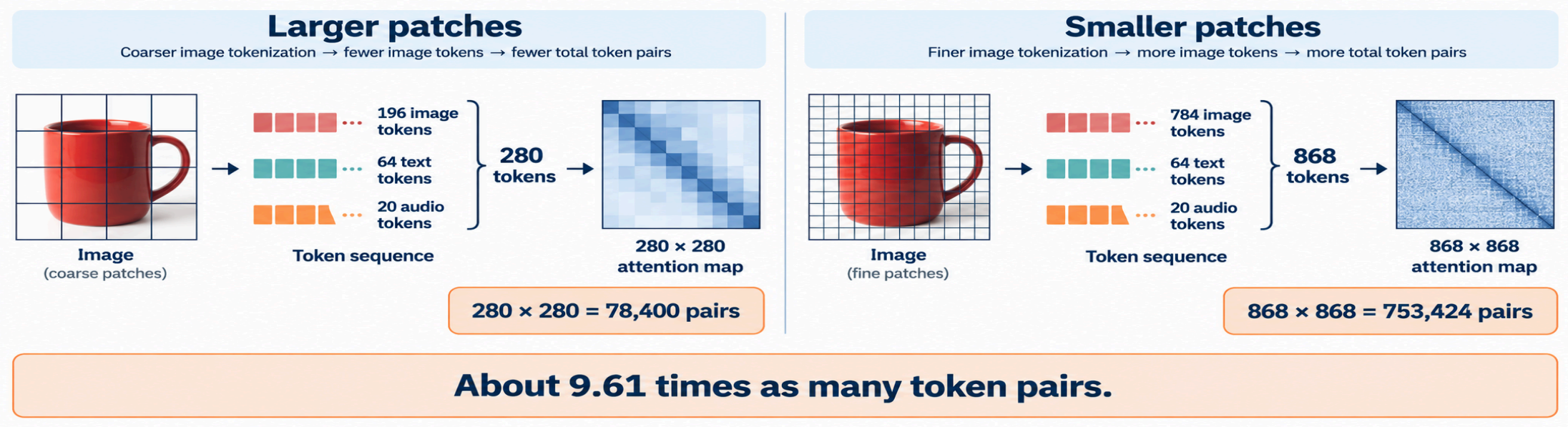
Without position information or another order-sensitive mechanism, unmasked self-attention is permutation-equivariant: reordering inputs reorders outputs rather than revealing the original order.

Position information may be added to embeddings or incorporated into attention. Later weeks cover modality-specific choices.

Check: If two patches have the same appearance, what tells the model which is on the left?

Source: Vaswani et al. - Attention Is All You Need, Section 3.5

Token Budget - A Worked Estimate



For a 224×224 image, 16×16 pixel patches produce $(224/16)^2 = 196$ image tokens.

Changing only the patch size to 8×8 gives **784 image tokens**. Dense self-attention considers L^2 query-key pairs per layer and head.

Recap - Why Attention?

A fixed-size summary asks one vector to retain evidence for many questions while **attention lets each query retrieve a task-dependent mixture of values.**

We can ask three questions about the same scene:

1. What color is the mug? *Inspect its surface.*
2. How many apples are there? *Compare both objects.*
3. What is behind the apples? *Examine their relation to the book.*

The image tokens stay the same but the query-dependent summary changes. Cross-modal queries can link phrases to visual evidence.

Like consulting an archive, retrieval depends on the question.

Source: Vaswani et al., Attention Is All You Need (2017)

Queries, Keys, and Values

Attention retrieves information using learned numerical comparisons.

- A query represents what an output position is seeking, keys represent how candidate information can be matched, and values contain the information to combine.
- Similarity scores are normalized into weights, and a weighted sum of values forms the retrieved representation.

These vectors are learned projections, not explicit human-readable questions and answers.

Self-attention draws them from the same sequence, while cross-attention uses another sequence as the source of keys and values. We can track their origins and dimensions to understand the direction of information flow.

- **Query:** what this position is looking for
- **Key:** what each candidate advertises
- **Value:** the information returned if selected

Source: Vaswani et al., Attention Is All You Need (2017)

Queries, Keys, and Values

For inputs X:

$$Q = XW_Q, \qquad K = XW_K, \qquad V = XW_V$$

The projections are learned; “query” and “key” are functional roles, not fixed semantic types.

If X contains n tokens of width d, the projections produce one query, key, and value per token. Queries and keys are optimized for matching; values carry the content to be combined. Keeping these roles separate lets the model learn *where to retrieve from* independently of *what to retrieve*.

Source: Vaswani et al., Attention Is All You Need (2017)

Scaled Dot-Product Attention

For one attention head, Q has one row per query and d_k columns; K has one row per source position and the same d_k columns.

Multiplying Q by K transposed produces one score for each query-source pair.

Division by the square root of d_k controls score scale, and row-wise softmax turns the scores into weights summing to one.

Multiplying those weights by V returns one weighted value vector per query.

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

Read it left to right:

1. compare every query with every key;
2. scale to keep logits numerically well behaved;
3. normalize each query's scores;
4. mix the values using those weights.

The score matrix is just an interaction map, not an explanation.

Each row of the softmax matrix sums to one and defines a weighted retrieval for one query. Dividing by $\sqrt{d_k}$ prevents dot products from growing with vector width and pushing softmax into nearly saturated regions where gradients become uninformative.

Why Scale by $\sqrt{d_k}$?

The scale factor is motivated by the size of a dot product as its dimension grows.

- If query and key components have approximately zero mean, unit variance, and suitable independence assumptions, the variance of their dot product grows with d_k.
- Large scores can make softmax extremely concentrated, leaving weak gradients for most alternatives.
- Dividing by the square root of d_k keeps the score variance more stable under those assumptions.

Without scaling, larger heads can produce very large logits. Softmax then becomes nearly one-hot, leaving little gradient signal for competing keys.

The factor $1/\sqrt{d_k}$ keeps the score scale more stable. It does not make attention interpretable. It makes optimization better behaved.

Walkthrough - Attention Retrieves Evidence

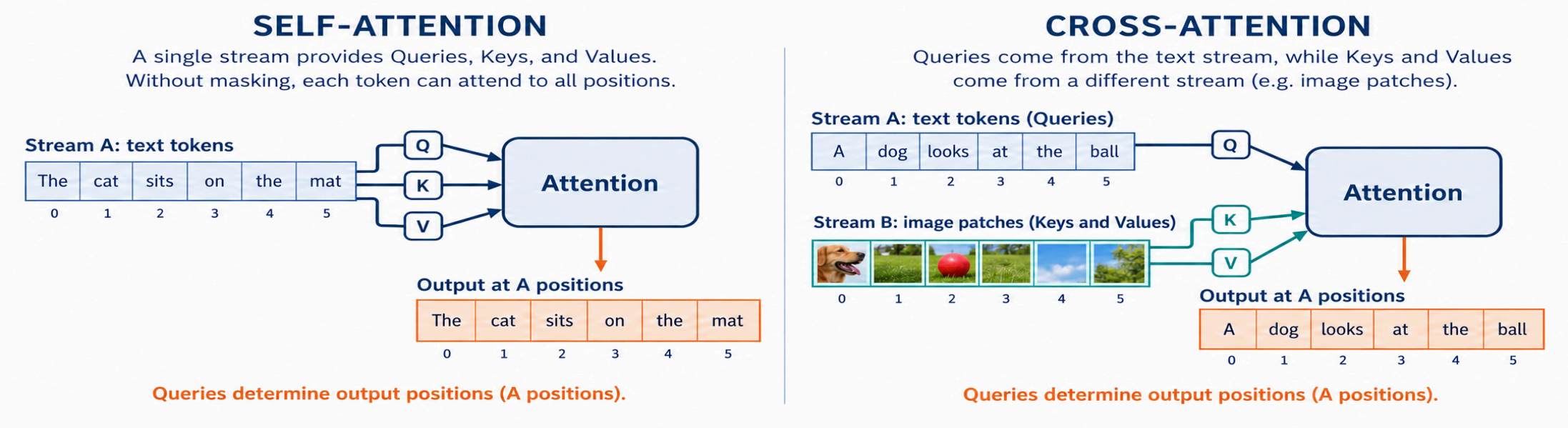
Suppose the text query is **“What color is the ball?”** and the image provides three patch tokens.

Candidate patch	Scaled score	Attention weight	Value contributes
red ball	2.0	0.80	red, round object
blue sky	0.2	0.13	blue background
green grass	−0.5	0.07	green texture

1. The query is compared with every patch key.
2. Softmax converts scores into relative weights.
3. The output is a weighted mixture of the patch values.
4. A later layer uses that mixture to predict **red**.

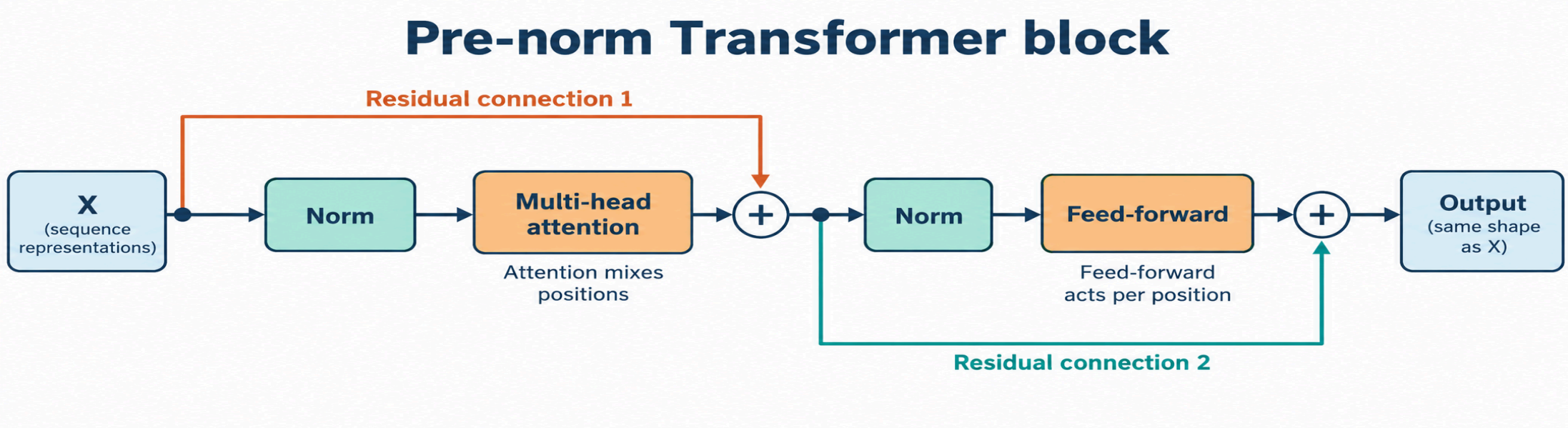
The mechanism enables evidence retrieval; training and evaluation determine whether useful retrieval is learned.

Self-Attention vs Cross-Attention



- **Self-attention:** Q, K, and V come from the same sequence.
- **Cross-attention:** Q comes from one sequence; K and V come from another.

The Transformer Block

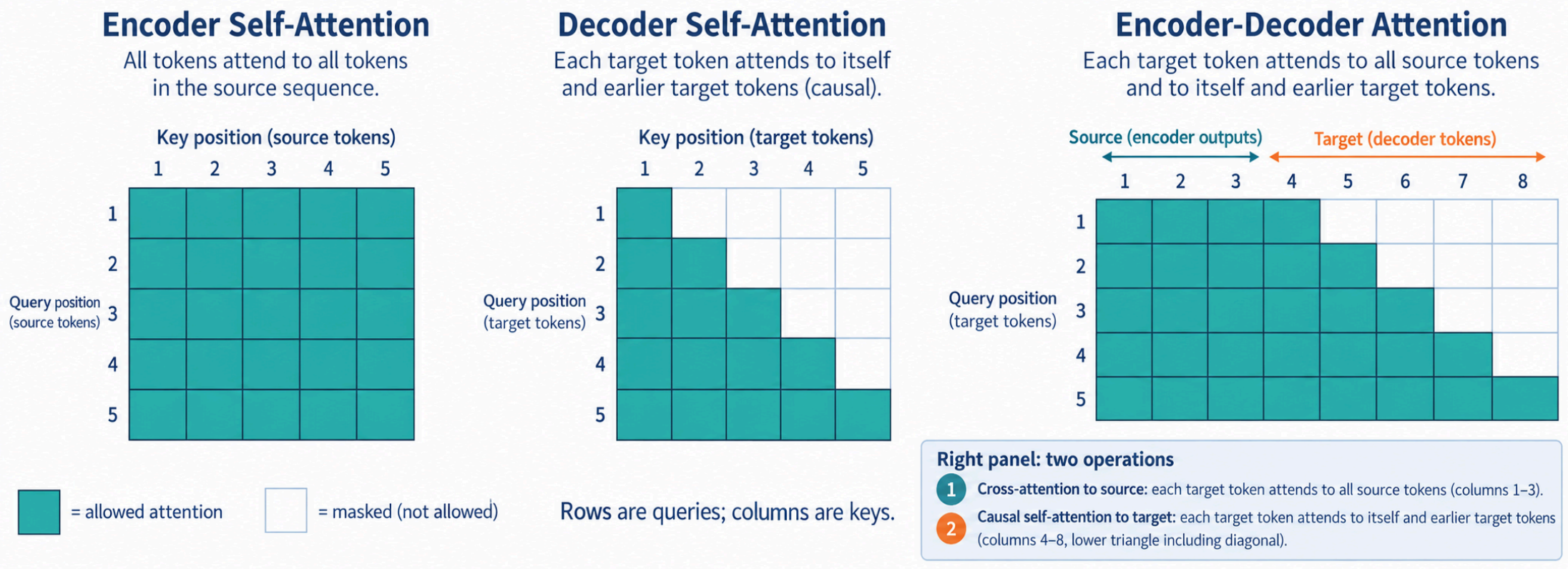


The **pre-norm** example applies normalization, multi-head attention, and a residual addition, then normalization, a feed-forward network, and another residual addition.

Attention mixes information **between positions**. The same feed-forward transformation acts **within each position**. Multiple heads learn different retrieval patterns; normalization regulates activation scale.

Residual paths add a learned update to the previous state and support deep optimization. They are additions, not concatenations. Other Transformer variants place normalization differently.

Three Transformer Families



- **Encoder-only:** bidirectional input attention, useful for representation, retrieval, and classification.
- **Decoder-only:** causal attention for sequential generation and instruction following.
- **Encoder-decoder:** source encoding plus causal target generation; target queries can also attend to the encoded source.

Block 5 - Interaction Needs Evidence

> What would distinguish useful cross-modal dependence from a shortcut or two independent predictions?

Complementarity and Redundancy Are Not the Same

Complementary information becomes useful when one source resolves an ambiguity that another cannot.

- A camera can identify which person moved their lips while a microphone supplies the words, allowing the system to associate speech with a speaker.

For an audiovisual speech sample, ask what happens when the audio becomes noisy:

Situation	Audio	Lip motion	What the combination provides
Clean speech	useful	useful	mostly redundant evidence
Background noise	unreliable	useful	complementary evidence
Dubbing or delay	misleading	useful	a conflict to diagnose

The relationship can change from sample to sample.

Complementarity and Redundancy Are Not the Same

Redundancy means that more than one observation carries some of the same task-relevant information.

- It can provide resilience when one source becomes noisy or unavailable.
- Both sound and lip movement may support recognition of a syllable, but their errors need not be independent.
- If both sources fail under the same condition, combining them may offer little protection.

But redundant high-dimensional tokens can also waste compute and amplify shared bias.

Redundancy is especially valuable under corruption or missing inputs: one source can substitute for another. However, repeated evidence is not independent evidence. Two captions copied from the same metadata field may agree perfectly while carrying the same error.

Conflict

Modalities can disagree because of:

- sensor error or latency;
- sarcasm and deliberate contradiction;
- distribution shift;
- adversarial manipulation;
- genuinely ambiguous evidence.

A useful system should represent uncertainty or ask for clarification; it should not average contradictions into confidence.

Conflict handling requires an explicit policy that defines how to trust the historically reliable source, estimate reliability per sample, maintain several hypotheses, or defer the decision.

Conflict Requires a Policy

Suppose a camera-based model reports **“road clear”**, while a lidar-based model reports **“obstacle close.”**

Possible policies include:

1. choose the historically more reliable sensor;
2. estimate reliability for this sample and weight the evidence;
3. keep both hypotheses and request another observation;
4. abstain because the cost of a collision is high.

The correct policy depends on the decision cost and the failure model. Averaging two contradictory scores is not a safety argument.

More Modalities Can Hurt

Adding modalities does not guarantee a better solution.

- More parameters and data alignment work
- Unequal learning speeds and modality dominance
- Missing inputs at deployment
- Spurious correlations unique to the training set
- Longer sequences and quadratic attention cost
- Larger privacy and attack surface

> Start with a useful baseline set: each modality alone, sensible combinations, and tests under corruption.

Additional modalities improve a system only when their useful information exceeds the estimation, synchronization, compute, and reliability costs they introduce.

Connections

A connection describes the relationship between elements across modalities:

- a phrase refers to a region;
- audio and lip motion share a time interval;
- a table cell supports a sentence;
- an action changes a later observation.

Connections may be known, weakly observed, latent, one-to-many, or temporally offset.

Connections answer **which elements could correspond or influence one another**.

- They may come from metadata, timestamps, spatial calibration, human annotation, or learned similarity.
- Incorrect connections contaminate supervision; missing connections force the model to infer correspondence jointly with the task.

Interactions

An interaction exists when using modalities together changes the computation needed for the task.

- additive evidence;
- redundancy and substitution;
- multiplicative or conditional effects;
- synergy where neither modality suffices alone;
- conflict requiring arbitration.

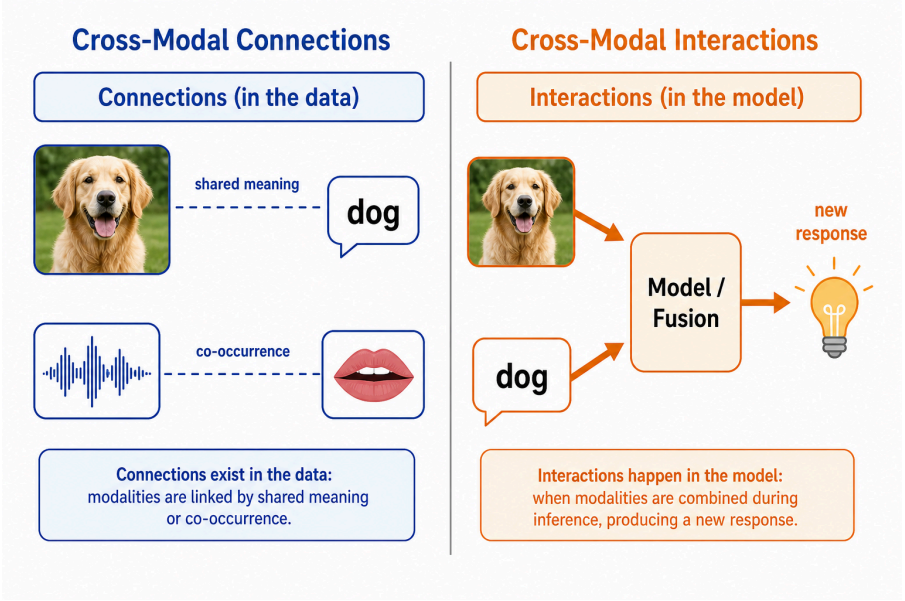
Interactions answer **how the joint presence of sources changes the prediction or internal computation**.

- Concatenating two inputs creates a connection path, but it does not prove that the model learned a conditional relationship between them.
- Interaction must be established through behavior under controlled changes.

Connections Are Not Interactions

A connection is an asserted correspondence, such as two recordings of the same event or a caption attached to an image. It tells the model which observations may be related, but does not establish that every element agrees.

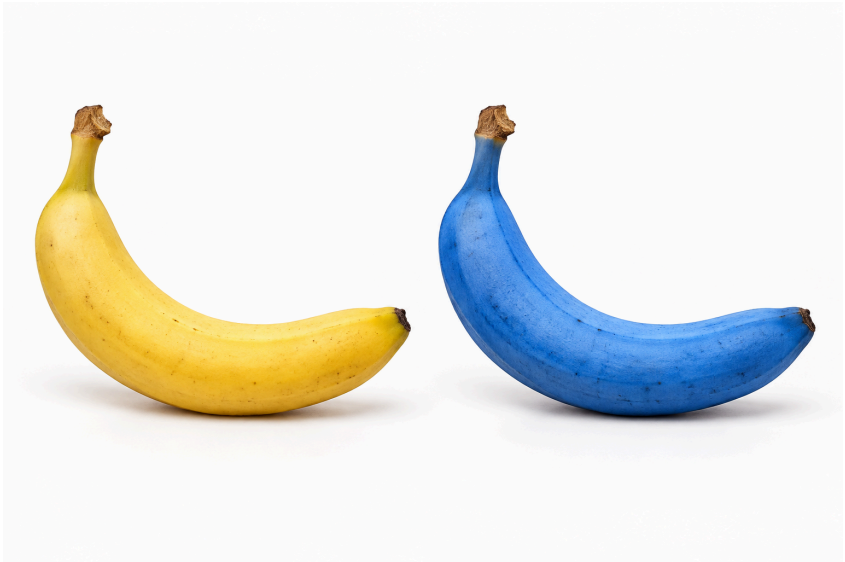
- A caption can omit most visible details, and a recording can begin late.
- Two inputs may be aligned yet only one drives the prediction.
- Two unaligned streams may still interact through a global decision.



Example - High Accuracy, Weak Multimodal Evidence

Imagine you have a dataset where most training questions asking “What color is the banana?” have the answer **yellow**.

Condition	Result	Interpretation to test
Full model, paired inputs	78%	Does the image contribute?
Question-only baseline	75%	Strong language shortcut?
Full model, shuffled test images	77%	Does correct pairing matter?
Full model, blue-banana input	“yellow”	Prior overrides this image



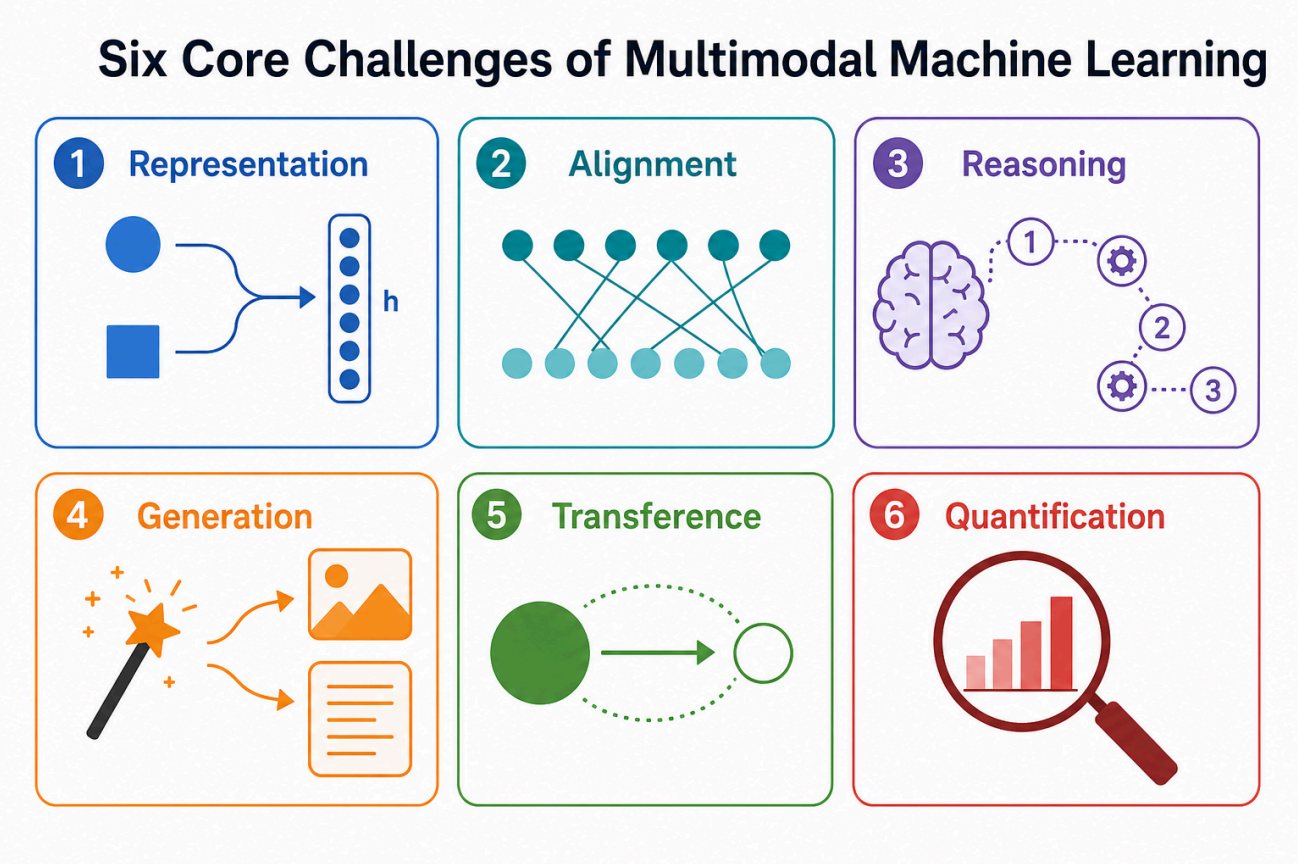
What Modality Tests Can and Cannot Show

Observation	Possible explanation	Follow-up control
Score falls after removal	Useful evidence was lost, or missing input is unfamiliar	Separately trained unimodal baseline and realistic missing-input evaluation
Score stays similar	Modality is ignored, redundant, or irrelevant on this subset	Test cases where that modality is necessary
Score falls after shuffling	Correct correspondence matters, or shuffling creates unrealistic inputs	Matched, plausible counterfactual pairs

Change relevant evidence while keeping the question and unrelated factors fixed. Predict the expected answer **before** running the test.

Input dependence under one test does not establish a particular interaction mechanism. Compare strong unimodal and simple-combination baselines, and examine failures as well as average scores.

Source: Hessel & Lee - Does my multimodal model learn cross-modal interactions? (2020)



A single design choice - such as visual tokenization - can affect representation, alignment, reasoning, generation, transfer, and evaluation simultaneously.

Source: Liang, Zadeh & Morency - Foundations and Trends in Multimodal Machine Learning

1 - Representation

How should heterogeneous observations become computational objects?

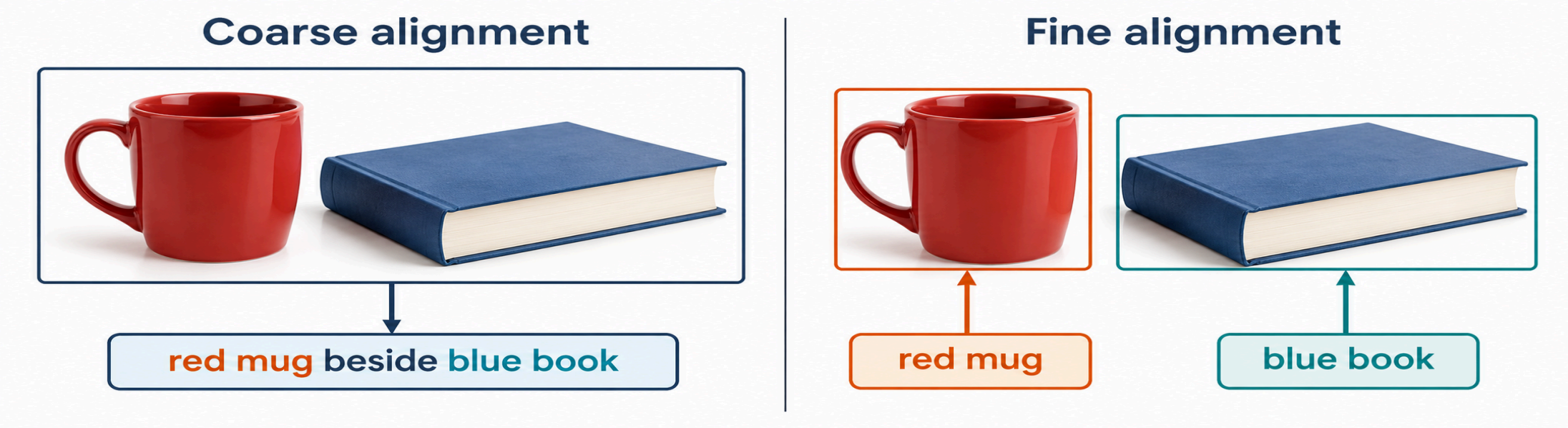
- continuous or discrete;
- shared or modality-specific;
- local tokens or global vectors;
- sufficient, compact, invariant, and recoverable - for which task?

The central trade-off is compression versus retained evidence.

- A global image vector is efficient for retrieval but may discard the spatial detail needed for grounded question answering.

A good representation exposes the information required by downstream operations without making computation unnecessarily expensive.

2 - Alignment



Coarse alignment says an image and caption belong together. **Fine alignment** links words to particular regions or moments. Image-level pairing alone does not supply every word-to-region label.

Other examples include speech-transcript timing, video event-action boundaries, and evidence-claim attribution in documents.

- Alignment may be supplied as data or supervision, learned in a shared space, or inferred at prediction time.

3 - Reasoning

How do we compose evidence across modalities over multiple steps?

Recognition asks “what is present?”

Reasoning asks “what follows from the relations among what is present?”

The danger here is that language priors can produce plausible answers without using the sensory evidence.

Multimodal reasoning therefore requires both **composition** and **grounding**.

- The system must combine intermediate facts while keeping them tied to observable evidence.
- Counterfactual inputs (*e.g.,* changing an object, relation, or temporal order) help reveal whether the answer follows the evidence or a familiar linguistic pattern.

4 - Generation

How do we create one or more modalities conditioned on others?

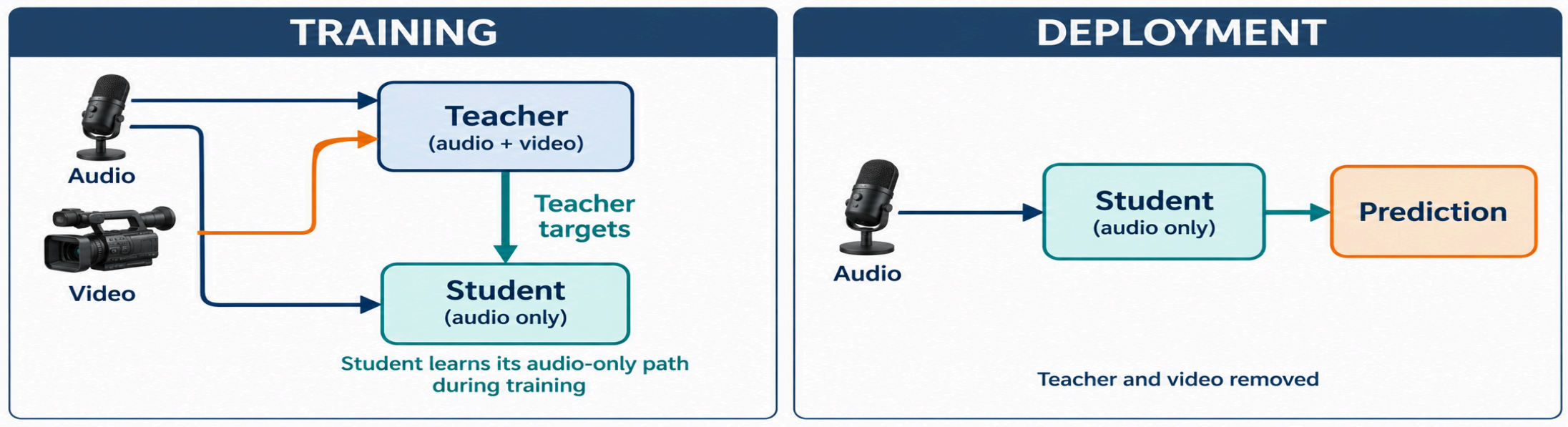
- captioning and speech synthesis;
- text-to-image, video, audio, motion, or 3D;
- editing and controllable generation;
- interleaved and any-to-any outputs.

Generation quality requires fidelity, conditioning alignment, diversity, and safety - not one score.

A fluent output can still ignore its conditioning input. Evaluation must separate perceptual quality from conditional faithfulness

- An attractive generated image is not a correct answer if it omits the requested object, changes identity, or violates spatial constraints.

5 - Transference



In **cross-modal distillation**, an audio-video teacher trains an audio-only student.

Related approaches use privileged training modalities, low-resource or multilingual transfer, missing-modality training, or small trainable connectors.

6 - Quantification

How do we measure multimodal information, interaction, capability, and risk?

- unimodal and shuffled-input baselines;
- grounding and counterfactual tests;
- statistical uncertainty and effect size;
- latency, memory, energy, and cost;
- benchmark validity and contamination.

Quantification is not optional. It determines what the field believes.

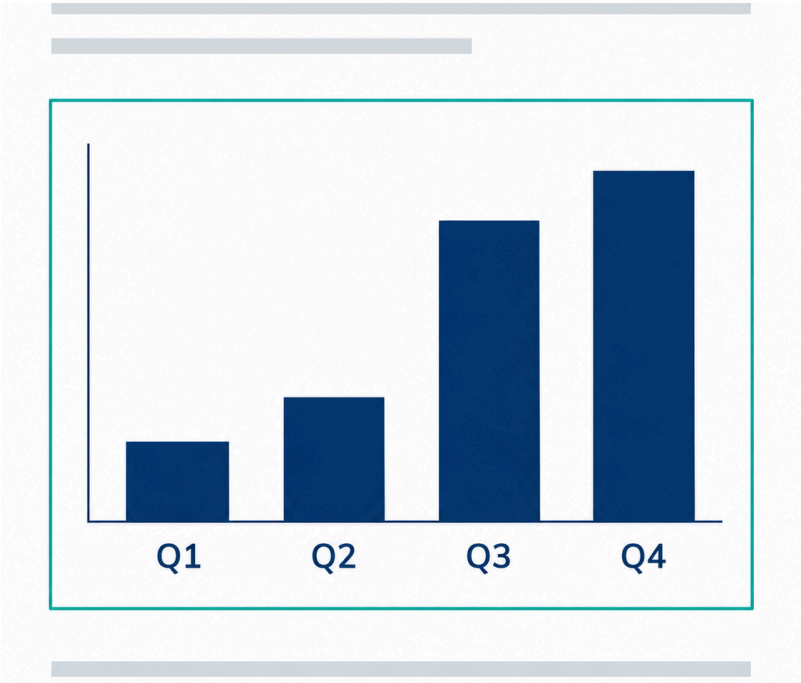
Every metric operationalizes a claim and omits something. A benchmark score should be accompanied by baselines, uncertainty, subgroup or condition analysis, and cost. When the evaluation does not isolate cross-modal dependence, it cannot support a strong claim about multimodal capability.

Walkthrough - Document QA from Page to Answer

Question: Which quarter has the largest increase over the previous quarter?

1. **Represent:** OCR supplies labels; visual patches preserve bar heights and layout.
2. **Connect:** Associate each quarter label with its bar.
3. **Compare:** Inspect Q2-Q1, Q3-Q2, and Q4-Q3. The largest increase is **Q3**, although Q4 is tallest.
4. **Test:** Edit the bars so a different quarter has the largest increase. Keep the question and labels fixed.

Compare image-plus-question with OCR-plus-question and question-only baselines. OCR alone may suffice when it preserves all required numerical values.



Key Points to Remember

1. A modality is an operational modeling distinction, not simply a file type.
2. Tokenization and representation determine the information available downstream.
3. Attention is learned retrieval; cross-attention is one fusion mechanism.
4. Connections do not guarantee interactions.
5. More modalities create costs and failure modes as well as information.
6. Every capability claim needs unimodal and counterfactual evidence.

Together these points form a reasoning chain:

- Representations determine available evidence; connections determine possible correspondence; interactions determine joint computation; and quantification determines whether our claims about that computation are justified.

End-of-Week Exercise

Pick a familiar AI system that claims to be multimodal.

What is the cheapest experiment that could reveal it is not actually using one of its modalities?

This is ungraded preparation. Bring a two-sentence answer next week: your intervention and the expected result if the model uses that modality. We will use several answers to open the representation lecture.

Thank You!

Contact Information

- **Email:** ekrem.cetinkaya@yildiz.edu.tr

- **Office Hours:** Wednesday 13:00 - 15:00 - Room C-120
- **Book a slot before coming to the office hours:** Booking Link
- **Course Repository:** GitHub Link

Remember: Start thinking about your project ideas!